1.8 Applications and Models

Name _____ Period ____ Date _____

Pre-Calculus 1.8 · Student sheet

I can…

- solve a right triangle.
- use angles of elevation and depression to solve real-life problems.
- use bearings to solve navigation problems.

Vocabulary · fill in the missing word

Solve a triangle — find all of its _____ sides and angles

Angle of depression — the angle from the _____ down to the line of sight

Bearing — a _____ given as an acute angle from north or south, like N 35° E

Part 1 · Measure the building

Height = distance · tan(angle) + eye height

Distance (ft)	Angle of elevation	Eye height (ft)	Height (ft)

Part 2 · Notes and examples (fill in each note, then work the example)

Solving a right triangle: the acute angles add to _____; use trig ratios for the sides.

Example 1

Solve right triangle ABC with C = 90°, A = 34.2°, and b = 19.4.

Example 2

A 110-foot ladder leans against a building at 72° with the ground. How high up the building does it reach?

Angle of depression from the top equals the angle of _____ from the bottom (alternate interior angles).

Example 3

From a 200-foot cliff, the angle of depression to a boat is 16°. How far is the boat from the base of the cliff?

Bearings: start at north or south, then turn _____ or west.

Example 4

A ship leaves port and sails 12 miles on a bearing of N 25° E. How far north and how far east is it?

Example 5

A ramp rises 3 feet over a horizontal run of 20 feet. What angle does it make with the ground?

Part 3 · Partner practice

Partner 1

An isosceles triangle has legs 12 and base angles 30°. Find its altitude to the base and the base.

Partner 2

A kite string is 20 meters long and makes a 70° angle with the ground. How high is the kite?